

Soo Wei Jie 苏伟捷

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EDUCATION

Shanghai Jiao Tong University

Shanghai, China

B.S.E. in Electrical and Computer Engineering; Minor in Data Science and Computer Science

Sep 2022 – Aug 2026

- **Relevant Coursework:** Machine Learning, Data Structures & Algorithms, Computer Organisation & Architecture, Probability & Statistics, Data Mining

EXPERIENCE

Robotics Reinforcement Learning Research Intern

Mar 2026 – Present

Shanghai Institute for Physical Artificial Intelligence and Robotics (PAIR)

Shanghai, China

- Built the lab's **concurrent** teleoperation and **demonstration-data collection pipeline** in **ManiSkill3 (SAPIEN/PhysX)**, streaming operator to drive a **Franka arm** fitted with multiple end-effectors (**Sharpa hand**, **Wuji hand**, **gripper**).
- Rendered the live view into a **VR headset** for closed-loop teleoperation; collected **demonstration data** to build training-ready datasets for **reinforcement learnings**, using dual control modes (controllers & hand tracking) to improve data consistency.
- Integrated the **Sharpa Manus SDK** and its **hand-retargeting algorithm**, fixing joint-axis mismatches and contact instabilities so teleoperated motion maps correctly onto each end-effector; also trained baseline **PPO** policies, iterating through evaluation to raise training efficiency by roughly **15%**.

Cloud Computing Engineering Intern

Oct 2025 – Jan 2026

ZStack Cloud, Shanghai Yunzhou Technology Co., Ltd

Shanghai, China

- Deployed IaaS on **Linux/KVM** across dual-management-node, SDS-converged, and SAN configurations, gaining end-to-end exposure to reproducible, large-scale compute and storage infrastructure.
- Diagnosed and resolved **3 infrastructure faults** on an unfamiliar stack and ran a full production server-health audit for an enterprise client.

Undergraduate Research Assistant

Sep 2023 – Aug 2024

Prof. Shane Johnson's Lab

Shanghai, China

- Develop a custom insole prototype with a pneumatic ergonomic system, implementing **embedded device control and sensor data acquisition** using **Raspberry Pi and Arduino**.
- Recruited 15 test subjects for functional testing and analysed 3D morphology and **force data** using **MATLAB and MeshLab**.

PROJECTS

<https://terny.ai> Agentic AI Platform, Research-Matching Agent

Dec 2025 – Feb 2026

- Architected and shipped a live **concurrent data pipeline (50+ users)** on a **FastAPI** backend, collecting and processing **150+ records** from **70+ sources in under 2 minutes** with multi-stage deduplication, filtering, and ranking.
- Built the retrieval layer with embedding-based semantic ranking and used **LangGraph** to orchestrate a multi-agent workflow, routing tasks across multiple model providers to balance latency, cost, and quality.
- Designed a **4-agent architecture** (3 specialised agents + an orchestrator routing by user intent) and built a **RAG chatbot** with grounded generation and **session persistence** for personalised matching and outreach.

Supervised Learning & Model Benchmarking Housing Price Prediction

Nov 2024

- Developed and **benchmarked** five supervised models with **cross-validation** and error analysis, reaching **0.365 RMSE** after feature engineering and hyperparameter tuning (top **10% of 40** in the class competition).

Out-of-Order RISC-V Processor (RV32IMA), Computer Architecture Capstone

Jan 2026 – Present

- Designed and verified a scalar **out-of-order RV32IMA** core in **SystemVerilog**; owned the Execution Engine (ALU, pipelined multiplier, and functional-unit to common-data-bus arbitration), tuning the datapath for throughput.

EXTRACURRICULAR & AWARDS

Vice Captain, University Varsity Football Team

Oct 2022 – May 2026

- Named **2025 SJTU Outstanding Student Athlete**; helped lead the varsity squad across training and competition. Also served as Vice President of the JI International Student Association.

Awards

- 2024 JI Encouragement Scholar; 2023 JI Development Scholar

SKILLS

Programming: Python, C++, C, SQL

Robotics / AI: demonstration-data collection, teleoperation, imitation learning (data), motion retargeting, reinforcement learning (PPO), PyTorch, ManiSkill3, SAPIEN/PhysX, Franka arm, LLMs, RAG, LangChain, LangGraph, multi-agent systems, agent orchestration, prompt engineering

Systems / Engineering: Linux, Docker, Git, data pipelines, reproducible training environments, TCP/socket programming, concurrent programming, CUDA, valgrind

Spoken Languages: English, Mandarin, Malay, Taiwanese